

Freight Load Rate Prediction

Spotter - Machine Learning Engineer assessment. Code, metrics and figures are reproducible with `python -m src.pipeline` (see README).

Summary

- **Task.** Predict posted_rate for 12,000 loads in Nov-Dec 2025 from 48,000 labelled loads in Jan-Oct 2025, and a daily December curve for one fixed lane.
- **Validation.** Rolling-origin time folds with a 2-month horizon (the real setup), plus a city-holdout and a random split for comparison. Every learned statistic is refitted inside each fold.
- **Model.** Two stages on log rate-per-mile: a LightGBM tree for the load-level price (lane/city/region encodings, distance, equipment, weight, quote_signal and its daily regime) plus a small ridge model for day-level effects (weekday, quarter-end ramp).
- **Result (final model A34).** Time-fold MAE \$40.4 (MAPE 1.85%), worst fold \$64.1; city holdout \$31.3, random split \$28.6; vs \$71.8 for ridge and \$70.1 for a plain LightGBM on all features.
- **Key finding.** quote_signal looked useless (pooled correlation ~0) but its relation to price flips sign between regimes that the daily mean quote identifies; modelling the regime is the largest single accuracy gain.

1. Data and key findings

- **Split in time.** Training covers 2025-01-01..10-31, validation 11-01..12-31: the task is a 2-month-ahead forecast. 8 cities appear only in validation (1,447 loads, 12.1%); every other validation load is on a lane seen in training (4,014 lanes, median 10 loads each).
- **Price structure.** Rate per mile falls with distance (~\$2.65 below 330 mi, ~\$1.89 above 2,200 mi); Reefer > Flatbed > Dry Van; heavier loads pay up to ~6% more.
- **market_index** is one value per day plus small per-load noise (within-day sd 0.025 vs 0.168 overall), with a strong weekly cycle (Thu high, Sun low). Its slow level is confounded with time: low in Jan-Feb when prices were low, low again in Aug-Oct when they were not.
- **Calendar.** Prices rise over the last ~3 weeks of each quarter (Mar, Jun, Sep) and drop after; weekday effect about +/-0.8%; a ~5% upward drift Jan-Oct not explained by the index. No visible public-holiday effect.
- **quote_signal regimes.** Pooled correlation with price is -0.04, so it first looked like noise, not leakage. Per week, on long loads, the correlation ranges -0.32..+0.49 and tracks the weekly mean quote ($r = 0.87$): positive when the mean quote is high (> 2.1: late Feb, Mar, Jun, Sep), negative when low (< 2.0: Apr, May, Jul, Oct), ~0 in between (Aug). The regimes cancel out when pooled. Nov-Dec sit at ~2.05, the transition zone.

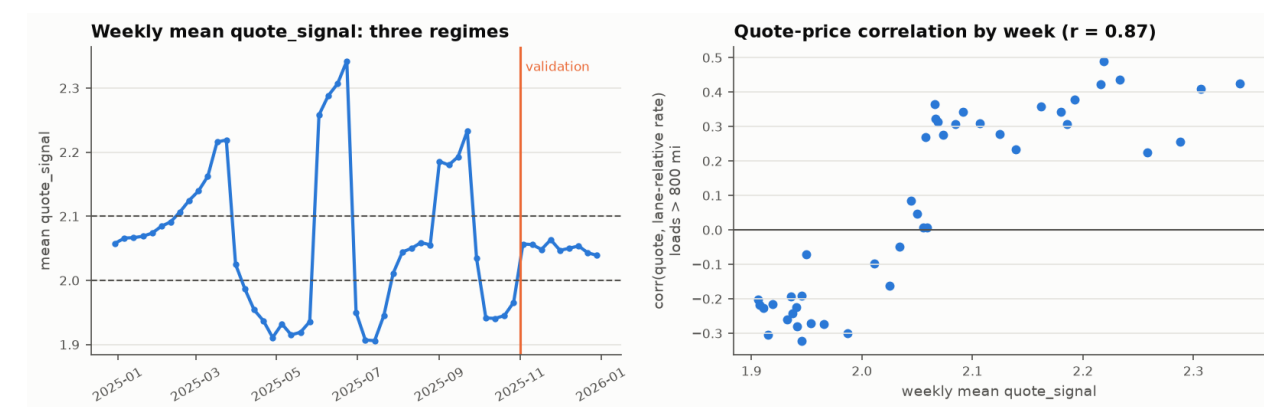


Figure 1. Left: weekly mean quote_signal forms three regimes. Right: the quote-price correlation per week follows the regime.

2. Data-quality issues and fixes

Validation rows are never dropped; feature problems are fixed or imputed and flagged. Only training rows with a corrupted target are removed. Fitted rules (weight medians, the corrupted-label filter) are refitted inside every CV fold.

Rule	Fix	Train rows	Valid rows
R1 city/equipment spelling	strip + title-case (defensive)	0	0
R2 negative weight	absolute value (sign flip)	292	145
R3 weight at cap/floor	keep value, add flag column	1,236	308
R4 missing weight	impute lane x equipment median (≥ 5 loads) else equipment median, fitted on training rows; flag	300	165
R5 missing market_index	same-date mean of the feature; flag	374	249
R6 clipped coordinates (kept)	keep; distance is authoritative	2,052	669
R7 corrupted posted_rate	drop from training (rate/expected outside 0.6-1.7)	677	0

- **Corrupted rates.** 677 training rates (1.4%) are x2-x6 away from the expected price, spread evenly over equipment, month, weekday, distance and city: random multiplicative errors. They are flagged by an out-of-fold robust (L1) GBM that uses no lane identity; a lane-median rule agreed on 673 rows but mis-flagged thin lanes (1-2 loads) where the bad row itself moves the median. Sanity check: Huber loss on uncleaned labels matches L2 on cleaned labels, plain L2 on uncleaned labels is much worse.
- **Coordinates** are distorted and partly clipped (e.g. Boston/Providence lon = -69.5); distance is consistent (~ 1.18 x great-circle, $\pm 5\%$ within lane), so distance is authoritative.
- **No duplicates**, unparseable dates or spelling variants were found (checked defensively).

3. Validation approach

- **Time folds (primary).** T1 Jan-Apr > May-Jun, T2 Jan-Jun > Jul-Aug, T3 Jan-Aug > Sep-Oct: a 2-month horizon after the training cut-off, mirroring Jan-Oct > Nov-Dec. The model is chosen on the worst fold, not only the mean.
- **City holdout.** 64 cities in 8 random groups; each fold holds out every load touching a group city as pickup OR delivery. It measures the unseen-city risk (12% of validation).
- **Random 5-fold** only as a contrast: train and test share dates, so it rewards memorising day-level prices; its MAE is 2.9x lower than the time folds for the first LightGBM and 1.4x lower for the final model.
- **Leakage guard.** Weight medians, the corrupted-label filter, target encodings and their priors, the unseen-city blanking (training rows only), the stage-2 model and the tree are refitted in every fold on that fold's training rows. Test rows get encodings from the training period only.

- **Labels.** Metrics are reported on clean labels and on raw labels (corrupted rows kept), since the scored labels likely contain the same ~1.4% corruption.

4. Model

- **Target:** log(rate per mile); predictions are $\exp(\cdot) \times \text{distance}$ (log vs rpm vs \$ targets were compared; smearing changed nothing measurable).
- **Stage 1 - LightGBM:** distance, great-circle distance, coordinates, equipment, weight (+ cap/floor/missing flags), per-load market-index deviation, quote_signal and the day's mean quote (regime), and out-of-fold smoothed target encodings of lane, pickup, delivery, 5-degree region and region pair. Unseen keys fall back to an equipment x distance-band prior; 12% of training rows get one city blanked so the model learns that case. Recency weights with a 60-day half-life.
- **Stage 2 - day effects:** ridge on the tree's date-grouped out-of-fold residuals: weekday dummies + a linear ramp over the last 21 days of a quarter. Keeping date features out of the tree stops it from memorising day-level prices that do not carry over to new dates.
- **Tested and rejected:** slow market-index level (-3..-5% bias two months out), holiday features (no gain; Thanksgiving/Christmas never seen), linear time trend (unstable: much worse on T2), level offsets from recent residuals (overshoot after the June peak), a ridge blend (weight tuned on T1-T2 = 0.60; failed on T3: \$44.5 vs \$26.8).

5. Results

Model	T1	T2	T3	Mean	Worst	MAPE %	City	Random
Baseline: global median \$/mile x distance	189.8	199.2	203.7	197.6	203.7	9.37	213.0	213.5
Baseline: lane median \$/mile	148.8	83.8	76.2	102.9	148.8	4.25	94.8	90.9
Baseline: ridge on log \$/mile	52.5	52.3	110.7	71.8	110.7	3.01	59.6	54.8
LightGBM, all features (first model)	59.2	70.0	81.2	70.1	81.2	3.11	26.6	23.8
Two-stage: tree + weekday/quarter-end model	55.9	80.2	30.9	55.7	80.2	2.38	42.9	41.0
+ coarse quote regime + recency (smooth alt.)	42.5	71.4	32.4	48.8	71.4	2.12	38.4	36.7
+ trailing quote x regime (smooth alt.)	32.8	68.2	30.2	43.7	68.2	1.95	28.4	26.3
FINAL (A34): + daily quote regime + recency	30.2	64.1	26.8	40.4	64.1	1.85	31.3	28.6

MAE in \$ on clean labels. T1-T3 = time folds; City = city-holdout mean; Random = random 5-fold mean.

- **Versus baselines:** mean time-fold MAE is 44% below ridge and 61% below the lane median, and lower than the first LightGBM on every fold. Ridge remains better on T2 (\$52.3 vs \$64.1) but fails on T3 (\$110.7, -5% bias when the market index is low).
- **Raw labels:** time-fold MAE \$94.3 and RMSE \$627 with corrupted rows kept (clean: \$40.4 / \$63). RMSE on raw labels is dominated by the ~1.4% corrupted rows for every model.
- **Unseen cities:** city-holdout MAE \$31.3 vs \$28.6 on the random split: the penalty for an unknown city is small (~\$3).
- **Where it errs:** the weakest fold is T2 (Aug under-predicted ~3%: August's quote regime was neutral while January, the only similar training month, was positive). MAPE is highest on short loads.

6. December chart

The December file has no coordinates, market_index or quote_signal. Coordinates are looked up per city (identical for every load of a city); market_index and the daily quote regime use the real daily means of validation.csv (input features, not targets; a forecast from October missed the index by 0.02-0.07); quote_signal is the median of comparable training loads (Dry Van, 300-420 mi: 2.108). December 2025 lies outside the training range, so the model contains no absolute-time feature a tree would have to extrapolate.

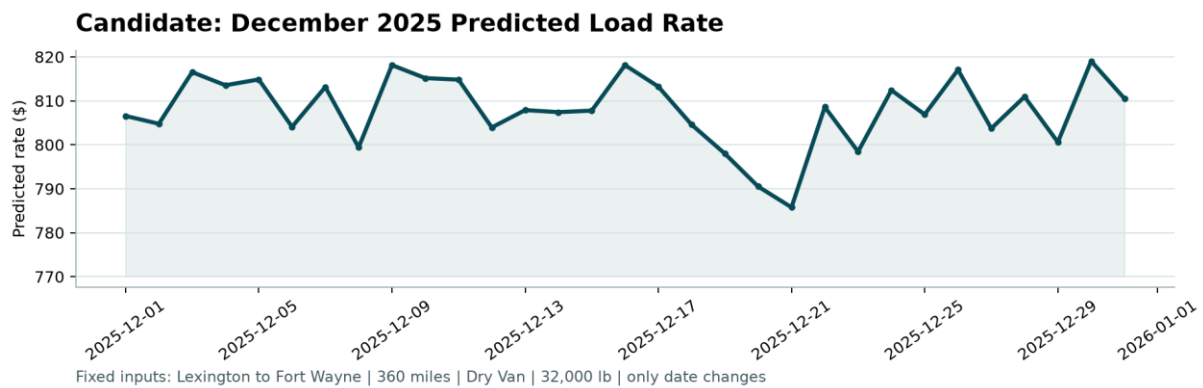


Figure 2. candidate_december.png produced by score.py.

- **Range** \$786-\$819 (mean \$808; recent Aug-Oct Dry Van loads on this lane scaled to 360 mi average ~\$808).
- **Stage-2 day effects** are small in the final model (Thursday +0.6% vs Monday, quarter-end ramp +0.2% by 31 Dec): the daily quote mean also carries day-level information, so the tree absorbs much of that movement.
- **Day-to-day jitter:** the load-level part moves between \$788 and \$816 because the daily mean quote (2.009-2.089) sits on the regime boundary, where the tree switches between the positive (Jan-like) and neutral (Aug-like) regime. Smoothed regime features (coarse bins, trailing 7/14-day means, quote x regime interaction) reduced or removed the jitter but cost 8-21% mean MAE on the time folds (a fully smooth weekday + quarter-end curve cost 21%); since validation accuracy is what is scored, the more accurate model was kept, and the same model produced both files.

7. Limitations and next steps

- **No Christmas/year-end period in training:** Thanksgiving and Christmas are predicted like ordinary days (e.g. 25 Dec as a Thursday).
- **Quarter-end effect** is learned from only three quarter-ends (Mar, Jun, Sep); the year-end may differ.
- **Quote regime ambiguity:** Nov-Dec sit at the regime boundary (~2.05), where training has only Jan (positive) and Aug (neutral) as examples; this is the largest source of risk and of the December jitter.
- **Price-level drift** is only partly predictable; level errors of 1-3% per month remain the main error source on the time folds.
- **Next steps:** a longer history (a previous December), an explicit regime model (e.g. hidden-Markov on the daily quote), quantile outputs for pricing ranges, and monitoring of the regime indicator.